ECON 2316 • MICROECONOMIC THEORY

# Part II in Formulas

The calculus behind Chapters 4–6 — every key equation, in plain English

Dr. Richeng Piao • Northeastern University

# First: meet the new symbols

Five symbols carry the consumer model. Here is what each one is really saying.

| | |
|---|---|
| $U(x_1, x_2)$ | **A utility function.** A score for each bundle — only the ranking matters, not the number itself. |
| $MU_i = \dfrac{\partial U}{\partial x_i}$ | **Marginal utility.** The next-unit move from Ch 1, now with a partial derivative because utility depends on two goods. |
| $\mathrm{MRS}$ | **Marginal rate of substitution.** The slope of an indifference curve — how much of good 2 the consumer would swap for one more unit of good 1. |
| $\lambda$ | **The Lagrange multiplier.** The shadow price of the budget — how many extra "utils" one more dollar of income would buy. |
| $h(p, \bar{u}),\ e(p, \bar{u})$ | **Hicksian demand & expenditure.** The cheapest way to hit a fixed utility level — the "dual" problem that powers the Slutsky equation. |

*Every formula in this deck is built from those five ideas, plus the calculus tools from Part I.*

NEW MOVE 1 OF 5

# Multivariable Utility — MU & MRS

Partial derivatives, but now they score “the next unit of this good.”

# Marginal utility — the partial derivative of $U$



$$MU_1 = \frac{\partial U(x_1, x_2)}{\partial x_1}, \qquad MU_2 = \frac{\partial U(x_1, x_2)}{\partial x_2}$$

**IN PLAIN ENGLISH**

Same idea as marginal cost from Ch 1 — the slope of a total. Utility now depends on two goods, so the partial derivative tells you the boost from one more unit of good $i$ **holding the other good fixed**.

*Diminishing $MU$ just means the curve flattens out: $\partial MU_i / \partial x_i < 0$. The third bite of pizza beats the eighth.*

# MRS — the slope of an indifference curve

MULTIVARIABLE UTILITY

Ch 4 • §4.3

$$\mathrm{MRS}_{1,2} = -\left.\frac{dx_2}{dx_1}\right|_{U=\bar{U}} = \frac{MU_1}{MU_2}$$

**IN PLAIN ENGLISH**

Walk along an indifference curve. To stay at the same utility, every extra unit of good 1 has to be paid for with some loss of good 2 — that trade-off rate is the MRS. **Implicit differentiation** of $U(x_1, x_2) = \bar{U}$ gives the clean ratio $MU_1/MU_2$.

*Convex preferences $\Rightarrow$ MRS falls as you move down the curve: the more you already have of good 1, the less of good 2 you'd give up for another unit.*

# Monotonic transformations — same preferences, same MRS

MULTIVARIABLE UTILITY

Ch 4 • §4.4

$$V = f\big(U(x_1, x_2)\big),\ f' > 0 \ \Rightarrow\ \mathrm{MRS}_V = \frac{f'(U)\,MU}{f'(U)\,MU}$$

**IN PLAIN ENGLISH**

Re-scale utility — take a log, add a constant, square it — and the **chain-rule factor cancels** from numerator and denominator of the MRS. Same indifference curves, same choices. Utility numbers are arbitrary; ratios of marginal utilities are not.

*This is the formal reason "utility is ordinal." The economics lives in the ranking, not in the level.*

NEW MOVE 2 OF 5

# The Four Canonical Utility Families

Different shapes of indifference curve, different stories about substitutability.

# One template, four shapes

UTILITY FAMILIES

Ch 4 • §4.5

$$U = x_1^{\alpha}\, x_2^{1-\alpha}$$

Cobb-Douglas — smooth tradeoff

$$U = a\,x_1 + b\,x_2$$

Perfect substitutes — constant MRS

$$U = \min\big(a\,x_1,\; b\,x_2\big)$$

Perfect complements — L-shaped, kink at the ridge

$$U = v(x_1) + x_2$$

Quasilinear — linear in good 2

IN PLAIN ENGLISH

Goods that swap smoothly use Cobb-Douglas; goods that are interchangeable (store-brand sugar) use perfect substitutes; goods that go together (left/right shoes) use perfect complements; one good with constant marginal value (extra dollar at the end of the month) is quasilinear. **Each family pins down a specific MRS — and a specific demand curve in Ch 5.**

THE SETUP — THE LIMIT THE CONSUMER CANNOT BREAK

# The Budget Constraint

A straight line in $(x_1, x_2)$ space — with two comparative statics already built in.

# The budget set — and a free comparative static

BUDGET CONSTRAINT



$$p_1 x_1 + p_2 x_2 \ \le\ I, \qquad \text{slope} = -\frac{p_1}{p_2}$$

Demand is homogeneous of degree zero: $x_i^*(\,kp,\ kI\,) = x_i^*(p, I)$

IN PLAIN ENGLISH

The budget line is the boundary the consumer cannot cross. A price change **rotates** it; an income change **shifts it parallel**. Scale prices and income by the same factor and nothing moves — only relative prices and real income matter.

*This is why economists write $p_1/p_2$, not $p_1$ alone. Inflation by itself is invisible to the model.*

NEW MOVE 3 OF 5

# Constrained Optimization — The Lagrangian

Part I's $\max f \text{ s.t. } g \leq 0$, now solved for two goods with one method.

# Tangency — two forms of the same rule

OPTIMIZATION

Ch 5 • §5.2

$$\text{MRS} = \frac{MU_1}{MU_2} = \frac{p_1}{p_2}$$

Slope form (the tangency condition)

$$\frac{MU_1}{p_1} = \frac{MU_2}{p_2}$$

Ratio form (equimarginal principle)

IN PLAIN ENGLISH

At the best bundle, the consumer's willingness to trade good 1 for good 2 **matches the market's price ratio**. Equivalently, the last dollar buys the same amount of extra utility no matter which good you spend it on — otherwise reallocate.

*Same idea as Part I's FOC: set the slope of the "goal" equal to the slope of the "limit."*

# The Lagrangian — and what $\lambda$ really is

OPTIMIZATION



$$\mathcal{L}(x_1, x_2, \lambda) \;=\; U(x_1, x_2) \;+\; \lambda\,\big(\,I - p_1 x_1 - p_2 x_2\,\big)$$

FOCs: $MU_i = \lambda\, p_i$ for each good $\Rightarrow \lambda = \dfrac{\partial V}{\partial I}$ (marginal utility of income)

**IN PLAIN ENGLISH**

The Lagrangian glues the goal and the limit into one expression. Take partial derivatives, set them to zero, and the tangency rule falls out automatically. The multiplier $\lambda$ is the **shadow price of the budget**: one more dollar of income would buy $\lambda$ extra utils.

*Same template extends to labor-leisure, time budgets, intertemporal choice — anywhere you maximize something subject to a linear constraint.*

# Solved templates — what the Lagrangian buys you

MARSHALLIAN DEMAND



$$x_1^* = \frac{\alpha\, I}{p_1}, \qquad x_2^* = \frac{(1-\alpha)\, I}{p_2}$$

Cobb-Douglas $U = x_1^{\alpha} x_2^{1-\alpha}$ — constant expenditure shares

$$v'(x_1^*) = \frac{p_1}{p_2} \quad \Rightarrow \quad \frac{\partial x_1^*}{\partial I} = 0$$

Quasilinear $U = v(x_1) + x_2$ — zero income effect on $x_1$

IN PLAIN ENGLISH

Cobb-Douglas spends a constant **fraction** of income on each good — double income, double quantities. Quasilinear pins $x_1^*$ **only to prices** — extra income flows into good 2 alone. Perfect complements ride the ridge $ax_1 = bx_2$; perfect substitutes pick the cheaper good (a corner).

*Memorize these closed forms — they show up in every Ch 5 walkthrough and again in Ch 6's Slutsky decomposition.*

NEW MOVE 4 OF 5

# Income & Substitution Effects

A price change has two channels. Pulling them apart is the Ch 6 punchline.

# Income elasticity — classifying a good

INCOME EFFECTS

Ch 6 • §6.1

$$\varepsilon_I = \frac{\partial x}{\partial I} \cdot \frac{I}{x}$$

Engel curve: $x = x^*(p, I)$ at fixed

**IN PLAIN ENGLISH**

Same template as the demand elasticity from Ch 3 — just with income on the bottom. The Engel curve traces out how purchases of one good move as income rises.

### READING THE SIGN OF $E_I$

| | | |
|---|---|---|
| $\varepsilon_I > 1$ — **luxury.** Demand rises faster than income. | $0 < \varepsilon_I < 1$ — **necessity.** Demand rises, but more slowly than income. | $\varepsilon_I < 0$ — **inferior.** Demand falls as income rises. |

# Decomposing a price change — two channels

INCOME & SUBSTITUTION

Ch 6 • §6.2

$$\underbrace{\Delta x_1}_{\text{total effect}} = \underbrace{\Delta x_1^{\text{sub}}}_{\text{substitution effect}} + \underbrace{\Delta x_1^{\text{inc}}}_{\text{income effect}}$$

**IN PLAIN ENGLISH**

When $p_1$ falls, two things happen at once. Good 1 is **relatively cheaper** — the consumer substitutes toward it (always). And the consumer's **real purchasing power rises** — for a normal good, that boosts demand further; for an inferior good, it pulls the other way.

*Sub effect on its own = move along the original indifference curve to the new price ratio. Income effect on its own = parallel shift of the budget to the new purchasing power.*

NEW MOVE 5A OF 5

# Duality & Hicksian Demand

The same optimum, approached from the other side — minimum spending for a fixed utility.

# Primal vs. dual — and Shephard's lemma

DUALITY

Ch 6 • §6.2.3

$$\max_x \; U(x) \;\; \text{s.t.} \;\; p \cdot x \le I$$

Primal — Marshallian $x^M(p,\, I)$

$$\min_x \; p \cdot x \;\; \text{s.t.} \;\; U(x) \ge \bar{u}$$

Dual — Hicksian $h(p,\, \bar{u})$

$$e(p,\, \bar{u}) \;=\; p \cdot h(p,\, \bar{u})$$

$$h_i(p,\, \bar{u}) \;=\; \frac{\partial e(p,\, \bar{u})}{\partial p_i}$$

IN PLAIN ENGLISH

Two questions about the same optimum: “maximize utility on a budget” and “hit a utility level for the least money.” They yield the same bundle. The dual gives you **Hicksian (compensated)** demand — the pure substitution response, with no income channel mixed in. Shephard says: to read the Hicksian, differentiate the cheapest-bundle cost with respect to that good's price.

NEW MOVE 5B OF 5 — THE SYNTHESIZING IDENTITY

# The Slutsky Equation

One line that ties Marshallian, Hicksian, income, and the sign chart together.

# The Slutsky equation — total = substitution + income

SLUTSKY EQUATION

Ch 6 • §6.3

$$\underbrace{\frac{\partial x_i^M}{\partial p_j}}_{\text{total (Marshallian)}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{substitution (Hicksian)}} - \underbrace{x_j^M \cdot \frac{\partial x_i^M}{\partial I}}_{\text{income channel}}$$

Own-price substitution is always non-positive: $\partial h_i / \partial p_i \le 0$

**IN PLAIN ENGLISH**

Differentiate the duality identity $x_i^M(p,\ I) = h_i\big(p,\ V(p, I)\big)$ and the Slutsky equation falls out. The Marshallian slope you observe in the market is the **pure substitution slope** (always away from the pricier good) **minus** the income channel, weighted by how much of good $j$ the consumer was buying.

*Quasilinear preferences have $\partial x_1 / \partial I = 0$, so the income term vanishes and Marshallian = Hicksian. That is why quasilinear is the workhorse for clean welfare math.*

# Sign chart — normal, inferior, Giffen

SLUTSKY — SIGN CLASSIFICATION



$$\frac{\partial x_1^M}{\partial p_1} = \underbrace{\frac{\partial h_1}{\partial p_1}}_{\leq 0 \text{ always}} - \underbrace{x_1^M \cdot \frac{\partial x_1^M}{\partial I}}_{\text{sign} = \text{sign}(\varepsilon_I)}$$

READING THE TABLE

**Normal good**

Sub $< 0$, income channel $< 0$. **Demand slopes down.**

**Inferior good — sub dominates**

Sub $< 0$, income channel $> 0$. **Demand still slopes down.**

**Giffen good — income dominates**

Sub $< 0$, but a large positive income channel **flips the sign — demand slopes up.**

Giffen is rare and requires (i) inferior and (ii) a very large income share — e.g. Jensen-Miller's Hunan rice.

# Five new moves. That is the whole consumer model.

| | |
|---|---|
| **Multivariable utility** | Partial derivatives in a new role: $MU_i = \partial U / \partial x_i$; slope of an indifference curve $\mathrm{MRS} = MU_1/MU_2$. |
| **Four canonical families** | Cobb-Douglas, perfect substitutes, perfect complements, quasilinear — each pins down a specific MRS and a specific demand. |
| **The Lagrangian** | Part I's $\max f \text{ s.t. } g \leq 0$, now mechanized. Tangency $\mathrm{MRS} = p_1/p_2$; $\lambda$ = shadow price of income. |
| **Income & substitution** | Two channels behind every price change. Income elasticity $\varepsilon_I$ classifies the good as normal, necessity, luxury, or inferior. |
| **Duality & the Slutsky equation** | Hicksian demand $h = \partial e / \partial p$ isolates the substitution channel. Slutsky stitches it back to the Marshallian demand the market actually shows you. |

**Every chapter from 7 onward** — production, costs, market power — reuses these moves with the names changed. Firms maximize profit subject to a production limit, just as consumers maximize utility subject to a budget. The economics changes; the calculus does not.